

STACK

Q9)

Question:

DS	Session	Stack	Question Information	Level 1	Challenge 49
Question description					
Rathik organized technical round interview in Macrosoft for the set of computer science candidates.					
The problem is to perform Implement a stack using single queue. you have to use queue data structure, the task is to implement stack using only given queue data structure.					
Rathik have given the deadline of only 5 minutes to complete the problem.					
Can you Help the candidates to complete the problem within the specified time limit ?					
Function Description					
x is the element to be pushed and s is stack					
push(s, x)					
1) Let size of q be s.					
2) Enqueue x to q					
3) One by one Dequeue s items from queue and enqueue them.					
Removes an item from stack					
pop(s)					
1) Dequeue an item from q					
Constraints					
$0 < n, m < N$					
$1 < arr[i] < 1000$					

PROGRAM:

```
1  #include <stdio.h>
2
3  int main() {
4      int n, m;
5      int q[1000], front = 0, rear = 0, size = 0;
6
7      scanf("%d %d", &n, &m);
8
9      for (int i = 0; i < n; i++) {
10         int x;
11         scanf("%d", &x);
12
13         q[rear++] = x;
14         size++;
15
16         for (int j = 0; j < size - 1; j++) {
17             q[rear] = q[front++];
18             rear++;
19         }
20     }
21
22     printf("top of element %d\n", q[front]);
23
24     for (int i = 0; i < m; i++)
25         front++;
26
27     printf("top of element %d", q[front]);
28
29     return 0;
30 }
```

OUTPUT:

TEST CASE 1:

```
5 2
1 2 3 4 5
top of element 5
top of element 3
```

TEST CASE 2:

```
15 4
9 8 7 6 5 5 6 7 8 3 11 2 3 4 5
top of element 5
top of element 11
```